
SIMON N. VANDEKAR

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June 18, 2025

Office Address:

Department of Biostatistics
Vanderbilt University Medical Center
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Office Phone Number:

615-875-2325

Education:

University of Pennsylvania, Philadelphia, PA 2014-2018
Ph.D. in Biostatistics
Dissertation advisor: Russell T. Shinohara

Pennsylvania State University, State College, PA 2005-2009
B.S. in Psychology-Neuroscience

Academic Appointments:

Associate Professor of Biostatistics 2024-present
Assistant Professor of Biostatistics 2018-2024
Director of Graduate Recruitment 2022-present
Department of Biostatistics
Vanderbilt University, School of Medicine

Professional Activities:

Intramural:

Chair, 20th Anniversary Celebration Committee 2023
Member, Comprehensive Exam Committee 2021-2024
Member, Department of Biostatistics Search Committee 2021-2023
Member, Department of Biostatistics Graduate Admissions Committee 2022-present
Member, Summer Internship for Underrepresented Undergraduates 2022-present

Extramural:

Program Chair Elect - Section on Statistics in Imaging Section,
American Statistical Association (ASA) 2020-2021
Program Chair - Section on Statistics in Imaging Section, ASA 2021-2022
Program Chair - Statistical Methods in Imaging Conference 2022 2021-2022
President - Middle TN Chapter, ASA 2022-2024

Chair/Host - Statistical Methods in Imaging Conference 2022	2021-2022
Committee Member - Statistical Methods in Imaging Conference 2023	2022-2023

Editorial Experience:

Reviewer:

Bioinformatics, Biometrics, Biostatistics, Human Brain Mapping, Journal of the American Medical Association Network Open, Journal of the American Medical Association Psychiatry, Journal of the American Medical Association Otolaryngology, Journal of the American Statistical Association, IEEE Journal of Biomedical Health Informatics, Journal of the Royal Statistical Society Series B, The Lancet Psychiatry, Neuroimage, Neuron, PLOS ONE, Psychoneuroendocrinology, IEEE Transactions on Medical Imaging, Statistical Methods in Medical Research, WIREs Computational Statistics

Associate Editor:

Biostatistics	2024-present
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Teaching Experience:

Didactic:

Teaching Assistant, Categorical and Survival Analysis	Spring 2016
Teaching Assistant, Linear Models and GLMs	Spring 2017
Teaching Assistant, Categorical and Survival	Spring 2018
Instructor, Substantially revised BIOS6311	Fall 2020-Fall 2023
Instructor, Summer Internship for Underrepresented Undergrads	Summer 2022

Research Supervision:

Nicholas Strayer (PhD Biostatistics 2019), Role: Committee Member
 Kavya Sharman (PhD Chemical and Physical Biology 2022), Role: Committee Member
 Cody Heiser (PhD Chemical and Physical Biology 2023), Role: Committee Member
 Harsimran Kaur (PhD Chemical and Physical Biology), Role: Committee Member
 Coleman Harris (PhD Biostatistics 2022), Role: Dissertation Advisor
 Kenneth Liao (MS Biostatistics, 2024), Role: Dissertation Advisor
 Kaidi Kang (PhD Biostatistics), Role: Dissertation Advisor
 Jiangmei (Ruby) Xiong (PhD Biostatistics), Role: Dissertation Advisor
 Megan Jones (PhD Biostatistics), Role: Dissertation Advisor
 Julia Sheffield, PhD (Assistant Professor of Psychiatry and Behavioral Sciences),
 Role: K23 Biostatistics Mentor
 Heather Ward, MD (Assistant Professor of Psychiatry and Behavioral Sciences),
 Role: K23 Biostatistics Mentor
 Xinyu Zhang (PhD Biostatistics), Role: Research Assistantship Supervisor
 Haozheng Xu (MS Biostatistics, 2024), Role: Dissertation Advisor

Membership:

American Statistical Association	2014-present
Eastern North America Region (ENAR) of the International Biometric Society	2015-present
Organization for Human Brain Mapping	2015-2016

Grant Support:

R01AG081655 (Darby)	09/01/2025-08/31/2029
NIMH	0.60 calendar months
Antisocial Behavior in Frontotemporal Dementia: Behavioral Phenotypes, Neural Markers, and Decision-Making Mechanisms	\$4,273,262 (total with indirect)
Role: Co- Investigator	
R01MH111599 (Corbett)	07/08/2024-04/30/2029
NIMH	0.60 calendar months
Examining stress and arousal across pubertal development in ASD	\$4,095,953 (total with indirect)
Role: Co-Investigator	
R01MH133843 (Vandekar)	07/01/2023-06/30/2028
NIMH	1.20 calendar months
Precision brain charts for imaging-genomics of schizophrenia and the psychosis spectrum	\$181,525 (total)
Role: Site Principal Investigator	
R01DK133415 (McKernan)	05/05/2023-02/29/2028
NIDDK	0.60 calendar months
Mechanistic-Based Treatment of Interstitial Cystitis/Bladder Pain Syndrome	\$686,309
Role: Co-Investigator	
R01EB036368 (O'Grady)	05/01/2024-04/30/2028
NINDS	1.20 calendar months
Multimodal, quantitative MRI in the lumbosacral spinal cord in progressive multiple sclerosis	\$1,877,326
Role: Co-Investigator	
R01EB036368 (Smith)	03/01/2024-02/29/2028
NIBIB	0.60 calendar months
Development of Susceptibility Weighted MRI for the Human Spinal Cord	\$686,309
Role: Co-Investigator	
P50MH132642-01 (Woodward)	04/03/2023 - 03/31/2028
NIMH	0.3 calendar months
Functional and behavioral dissection of higher order thalamocortical circuits in schizophrenia	\$2,162,500 (total)
Role: Biostatistician	

R01MH112847-06A1 (Vandekar)	09/01/2022-06/30/2027
NIMH	1.2 calendar months
Inter-modal Coupling Image Analytics	\$26,855 (annual total)
Role: Site Principal Investigator	
 R01HD107695 (Corbett)	 09/01/2022 - 06/30/2027
NICHD	0.60 calendar months
Pubertal, Psychological and Physiological Development in Females with Autism	\$3,330,139
Role: Co-Investigator	
 K24MH126280 (Woodward)	 09/15/2021-08/31/2026
NIMH	0.3 calendar months
Research and mentoring on cognitive neuroscience and high-field MRI of the thalamus in schizophrenia	\$149,320 (annual total)
Role: Co-Investigator	
 R01NS117816 (Smith)	 02/15/2021-01/31/2026
NINDS	0.60 calendar months
Evaluating Advanced Diffusion of the Human Spinal Cord: Application to MS	\$1,068,430 (total)
Role: Co-Investigator	
 R01MH123563 (Vandekar)	 08/01/2020-05/31/2026
NIMH	2.40 calendar months
Semiparametric Inference for Psychiatric Neuroimaging	\$2,113,210 (total)
Role: Principal Investigator	
 R01MH070560 (Heckers)	 02/07/2005-03/31/2026
NIMH	1.2 calendar months
Imaging Hippocampal Function in Psychosis	\$818,430 (annual total)
Role: Co-Investigator	
 R01MH123662 (Taylor)	 09/01/2020 - 06/30/2025
NIMH	0.60 calendar months
Expansion of the Dopaminergic Dysfunction in Late-Life Depression Study (The D3 Study)	\$7,401,771 (total)
Role: Co-Investigator	
 R33MH120149 (Corbett)	 09/01/2020 - 03/31/2025
NIMH	0.60 calendar months
Enhancing Social Competence in Adults with Autism Spectrum Disorder: A Pilot RCT	\$1,436,787

Role: Co-Investigator

K23DK118118 (McKernan) 05/13/2019-03/31/2023
0.60 calendar months
NIDDK \$686,309
Optimizing Psychosocial Treatment of Interstitial Cystitis/Bladder Pain Syndrome
Role: Biostatistician

R01MH118273 (Heckers) 08/01/2019-05/31/2024
0.60 calendar months
NIMH \$3,689,156
The General Factor of Psychopathology in Psychosis and Severe Mental Illness
Role: Co-Investigator

R01NS109114 (Smith) 09/15/2018-06/30/2023
0.96 calendar months
NINDS \$1,733,700
Structural and functional MRI of the cervical spinal cord in multiple sclerosis
Role: Co-Investigator

U2CCA233291-01 (Coffey) 09/20/2018-06/30/2023
3.00 calendar months
NCI \$10,993,061
Integrative single-cell atlas of host and microenvironment in colorectal neoplastic transformation: data analysis
Role: Co-Investigator

R01MH115000 (Woodward) 06/04/2018-03/31/2022
0.60 calendar months
NIMH \$1,587,494
Development of thalamocortical circuits and cognitive function in healthy individuals and youth at-risk for psychosis
Role: Co-Investigator

Peer Reviewed Publications ([Google Scholar](#)):

1. Dennis, Nancy A., Caitlin R. Bowman, and Simon N. **Vandekar**. 2012. "True and Phantom Recollection: An fMRI Investigation of Similar and Distinct Neural Correlates and Connectivity." *Neuroimage* 59 (3): 2982–93.
2. Satterthwaite, Theodore D., Russell T. Shinohara, Daniel H. Wolf, Ryan D. Hopson, Mark A. Elliott, Simon N. **Vandekar**, Kosha Ruparel, Monica E. Calkins, David R. Roalf, and Efstathios D. Gennatas. 2014. "Impact of Puberty on the Evolution of Cerebral Perfusion during Adolescence." *Proceedings of the National Academy of Sciences*, 201400178.
3. Satterthwaite, Theodore D., Simon **Vandekar**, Daniel H. Wolf, Kosha Ruparel, David R. Roalf, Chad Jackson, Mark A. Elliott, et al. 2014. "Sex Differences in the Effect of Puberty on Hippocampal Morphology." *Journal of the American*

- Academy of Child and Adolescent Psychiatry 53 (3): 341–50.e1.
<https://doi.org/10.1016/j.jaac.2013.12.002>.
4. Schmitt, J. E., J. J. Yi, D. R. Roalf, L. A. Loevner, K. Ruparel, D. Whinna, M. C. Souders, et al. 2014. “Incidental Radiologic Findings in the 22q11.2 Deletion Syndrome.” *American Journal of Neuroradiology* 35 (11): 2186–91.
<https://doi.org/10.3174/ajnr.A4003>.
 5. Sun, Wenguang, Brian J. Reich, T. Tony Cai, Michele Guindani, and Armin Schwartzman. 2015. “Visual Exploration Differences During Relational Memory Encoding in Early Psychosis.” *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 77 (1): 112910.
 6. **Vandekar**, Simon N., Russell T. Shinohara, Armin Raznahan, David R. Roalf, Michelle Ross, Nicholas DeLeo, Kosha Ruparel, et al. 2015. “Topologically Dissociable Patterns of Development of the Human Cerebral Cortex.” *Journal of Neuroscience* 35 (2): 599–609. <https://doi.org/10.1523/JNEUROSCI.3628-14.2015>.
 7. Schmitt, J. Eric, Simon **Vandekar**, James Yi, Monica E. Calkins, Kosha Ruparel, David R. Roalf, Daneen Whinna, et al. 2015. “Aberrant Cortical Morphometry in the 22q11.2 Deletion Syndrome.” *Biological Psychiatry, Autism and Brain Development*, 78 (2): 135–43. <https://doi.org/10.1016/j.biopsych.2014.10.025>.
 8. Satterthwaite, Theodore D., Daniel H. Wolf, David R. Roalf, Kosha Ruparel, Guray Erus, Simon **Vandekar**, Efstathios D. Gennatas, et al. 2015. “Linked Sex Differences in Cognition and Functional Connectivity in Youth.” *Cerebral Cortex* 25 (9): 2383–94. <https://doi.org/10.1093/cercor/bhu036>.
 9. Satterthwaite, T. D., S. N. **Vandekar**, D. H. Wolf, D. S. Bassett, K. Ruparel, Z. Shehzad, R. C. Craddock, et al. 2015. “Connectome-Wide Network Analysis of Youth with Psychosis-Spectrum Symptoms.” *Molecular Psychiatry* 20 (12): 1508–15. <https://doi.org/10.1038/mp.2015.66>.
 10. Roalf, David R., Megan Quarmley, Mark A. Elliott, Theodore D. Satterthwaite, Simon N. **Vandekar**, Kosha Ruparel, Efstathios D. Gennatas, et al. 2016. “The Impact of Quality Assurance Assessment on Diffusion Tensor Imaging Outcomes in a Large-Scale Population-Based Cohort.” *NeuroImage* 125 (January): 903–19. <https://doi.org/10.1016/j.neuroimage.2015.10.068>.
 11. Satterthwaite, Theodore D., Daniel H. Wolf, Monica E. Calkins, Simon N. **Vandekar**, Guray Erus, Kosha Ruparel, David R. Roalf, et al. 2016. “Structural Brain Abnormalities in Youth With Psychosis Spectrum Symptoms.” *JAMA Psychiatry* 73 (5): 515–24. <https://doi.org/10.1001/jamapsychiatry.2015.3463>.
 12. Shanmugan, Sheila, Daniel H. Wolf, Monica E. Calkins, Tyler M. Moore, Kosha Ruparel, Ryan D. Hopson, Simon N. **Vandekar**, et al. 2016. “Common and Dissociable Mechanisms of Executive System Dysfunction Across Psychiatric Disorders in Youth.” *American Journal of Psychiatry* 173 (5): 517–26.
<https://doi.org/10.1176/appi.ajp.2015.15060725>.
 13. **Vandekar**, Simon N., Russell T. Shinohara, Armin Raznahan, Ryan D. Hopson, David R. Roalf, Kosha Ruparel, Ruben C. Gur, Raquel E. Gur, and Theodore D. Satterthwaite. 2016. “Subject-Level Measurement of Local Cortical Coupling.” *NeuroImage* 133 (June): 88–97.
<https://doi.org/10.1016/j.neuroimage.2016.03.002>.
 14. Satterthwaite, T. D., P. A. Cook, S. E. Bruce, C. Conway, E. Mikkelsen, E. Satchell, S. N. **Vandekar**, T. Durbin, R. T. Shinohara, and Y. I. Sheline. 2016.

- “Dimensional Depression Severity in Women with Major Depression and Post-Traumatic Stress Disorder Correlates with Fronto-Amygdalar Hypoconnectivity.” *Molecular Psychiatry* 21 (7): 894–902.
<https://doi.org/10.1038/mp.2015.149>.
15. Kaczurkin, Antonia N., Tyler M. Moore, Kosha Ruparel, Rastko Ciric, Monica E. Calkins, Russell T. Shinohara, Mark A. Elliott, et al. 2016. “Elevated Amygdala Perfusion Mediates Developmental Sex Differences in Trait Anxiety.” *Biological Psychiatry, Stress, Fear, and Anxiety*, 80 (10): 775–85.
<https://doi.org/10.1016/j.biopsych.2016.04.021>.
 16. Baum, Graham L., Rastko Ciric, David R. Roalf, Richard F. Betzel, Tyler M. Moore, Russell T. Shinohara, Ari E. Kahn, Simon N. **Vandekar**, Petra E. Rupert, and Megan Quarmley. 2017. “Modular Segregation of Structural Brain Networks Supports the Development of Executive Function in Youth.” *Current Biology* 27 (11): 1561–72.
 17. Roalf, David R., J. Eric Schmitt, Simon N. **Vandekar**, Theodore D. Satterthwaite, Russell T. Shinohara, Kosha Ruparel, Mark A. Elliott, Karthik Prabhakaran, Donna M. McDonald-McGinn, and Elaine H. Zackai. 2017. “White Matter Microstructural Deficits in 22q11. 2 Deletion Syndrome.” *Psychiatry Research: Neuroimaging* 268: 35–44.
 18. Sharma, Anup, Daniel H. Wolf, Rastko Ciric, Joseph W. Kable, Tyler M. Moore, Simon N. **Vandekar**, Natalie Katchmar, Aylin Daldal, Kosha Ruparel, and Christos Davatzikos. 2017. “Common Dimensional Reward Deficits across Mood and Psychotic Disorders: A Connectome-Wide Association Study.” *American Journal of Psychiatry* 174 (7): 657–66.
 19. **Vandekar**, Simon N., Haochang Shou, Theodore D. Satterthwaite, Russell T. Shinohara, Alison K. Merikangas, David R. Roalf, Kosha Ruparel, Adon Rosen, Efsthios D. Gennatas, and Mark A. Elliott. 2017. “Sex Differences in Estimated Brain Metabolism in Relation to Body Growth through Adolescence.” *Journal of Cerebral Blood Flow & Metabolism*, 39(3): 524–35. <https://doi.org/10.1177/0271678X17737692>.
 20. Reich, Brian J., Joseph Guinness, Simon N. **Vandekar**, Russell T. Shinohara, and Ana-Maria Staicu. 2018. “Fast Function-on-Scalar Regression with Penalized Basis Expansions.” *International Journal of Biostatistics* 74 (2): Article 28.
 21. Valcarcel, Alessandra M., Kristin A. Linn, Fariha Khalid, Simon N. **Vandekar**, Shahamat Tauhid, Theodore D. Satterthwaite, John Muschelli, Melissa Lynne Martin, Rohit Bakshi, and Russell T. Shinohara. 2018. “A Dual Modeling Approach to Automatic Segmentation of Cerebral T2 Hyperintensities and T1 Black Holes in Multiple Sclerosis.” *NeuroImage: Clinical* 20: 1211–21.
 22. Valcarcel, Alessandra M., Kristin A. Linn, Simon N. **Vandekar**, Theodore D. Satterthwaite, John Muschelli, Peter A. Calabresi, Dzung L. Pham, Melissa Lynne Martin, and Russell T. Shinohara. 2018. “MIMoSA: An Automated Method for Intermodal Segmentation Analysis of Multiple Sclerosis Brain Lesions.” *Journal of Neuroimaging* 28 (4): 389–98.
<https://doi.org/10.1111/jon.12506>.
 23. Reardon, P. K., Jakob Seidlitz, Simon **Vandekar**, Siyuan Liu, Raihaan Patel, Min Tae M. Park, Aaron Alexander-Bloch, Liv S. Clasen, Jonathan D. Blumenthal, and Francois M. Lalonde. 2018. “Normative Brain Size Variation and Brain

- Shape Diversity in Humans.” *Science* 360 (6394): 1222–27.
<https://doi.org/10.1126/science.aar2578>.
24. Xia, Cedric Huchuan, Zongming Ma, Rastko Ciric, Shi Gu, Richard F. Betzel, Antonia N. Kaczkurkin, Monica E. Calkins, et al. 2018. “Linked Dimensions of Psychopathology and Connectivity in Functional Brain Networks.” *Nature Communications* 9 (1): 3003. <https://doi.org/10.1038/s41467-018-05317-y>.
 25. Alexander-Bloch, Aaron F., Haochang Shou, Siyuan Liu, Theodore D. Satterthwaite, David C. Glahn, Russell T. Shinohara, Simon N. **Vandekar**, and Armin Raznahan. 2018. “On Testing for Spatial Correspondence between Maps of Human Brain Structure and Function.” *NeuroImage* 178 (September): 540–51. <https://doi.org/10.1016/j.neuroimage.2018.05.070>.
 26. **Vandekar**, Simon N., Theodore D. Satterthwaite, Adon Rosen, Rastko Ciric, David R. Roalf, Kosha Ruparel, Ruben C. Gur, Raquel E. Gur, and Russell T. Shinohara. 2018. “Faster Family-Wise Error Control for Neuroimaging with a Parametric Bootstrap.” *Biostatistics* 19 (4): 497–513. <https://doi.org/10.1093/biostatistics/kxx051>.
 27. **Vandekar**, Simon N., Theodore D. Satterthwaite, Cedric H. Xia, Azeez Adebimpe, Kosha Ruparel, Ruben C. Gur, Raquel E. Gur, and Russell T. Shinohara. 2019. “Robust Spatial Extent Inference with a Semiparametric Bootstrap Joint Inference Procedure.” *Biometrics* 75 (4): 1145–55. <https://doi.org/10.1111/biom.13114>.
 28. Corbett, Blythe A., Sara Ioannou, Alexandra P. Key, Catherine Coke, Rachael Muscatello, Simon **Vandekar**, and Ian Muse. 2019. “Treatment Effects in Social Cognition and Behavior Following a Theater-Based Intervention for Youth with Autism.” *Developmental Neuropsychology* 44 (7): 481–94. <https://doi.org/10.1080/87565641.2019.1676244>.
 29. McHugo, Maureen, Pratik Talati, Kristan Armstrong, Simon N. **Vandekar**, Jennifer Urbano Blackford, Neil D. Woodward, and Stephan Heckers. 2019. “Hyperactivity and Reduced Activation of Anterior Hippocampus in Early Psychosis.” *American Journal of Psychiatry*, October. <https://doi.org/10.1176/appi.ajp.2019.19020151>.
 30. Corbett, Blythe A., Simon **Vandekar**, Rachael A. Muscatello, and Yasas Tanguturi. 2020. “Pubertal Timing during Early Adolescence: Advanced Pubertal Onset in Females with Autism Spectrum Disorder.” *Autism Research* 13 (12): 2202–2215. <https://doi.org/10.1002/aur.2406>.
 31. Parikh, Meghana P., Simon **Vandekar**, Christina Moore, Linda Thomas, Nathan Britt, Bhinnata Piya, Laura S. Stewart, Einas Batarseh, Lubna Hamdan, and Steffany J. Cavallo. 2020. “Temporal and Genotypic Associations of Sporadic Norovirus Gastroenteritis and Reported Norovirus Outbreaks in Middle Tennessee, 2012–2016.” *Clinical Infectious Diseases* 71 (9): 2398–2404.
 32. Larsen, Bart, Josiane Bourque, Tyler M. Moore, Azeez Adebimpe, Monica E. Calkins, Mark A. Elliott, Ruben C. Gur, et al. 2020. “Longitudinal Development of Brain Iron Is Linked to Cognition in Youth.” *Journal of Neuroscience* 40 (9): 1810–18. <https://doi.org/10.1523/JNEUROSCI.2434-19.2020>.
 33. **Vandekar**, Simon, Ran Tao, and Jeffrey Blume. 2020. “A Robust Effect Size Index.” *Psychometrika* 85 (1): 232–46. <https://doi.org/10.1007/s11336-020-09698-2>.
 34. Alexander-Bloch, Aaron F., Armin Raznahan, Simon N. **Vandekar**, Jakob Seidlitz, Zhixin Lu, Samuel R. Matthias, Emma Knowles, et al. 2020. “Imaging Local

- Genetic Influences on Cortical Folding.” *Proceedings of the National Academy of Sciences* 117 (13): 7430–36. <https://doi.org/10.1073/pnas.1912064117>.
35. Avery, Suzanne N., Maureen McHugo, Kristan Armstrong, Jennifer U. Blackford, Simon **Vandekar**, Neil D. Woodward, and Stephan Heckers. 2020. “Habituation during Encoding: A New Approach to the Evaluation of Memory Deficits in Schizophrenia.” *Schizophrenia Research*, July. <https://doi.org/10.1016/j.schres.2020.07.007>.
 36. Hamdan, Lubna, Simon **Vandekar**, Andrew J. Spieker, Herdi Rahman, Danielle Ndi, Emily S. Shekarabi, Jyotsna Thota, et al. 2020. “Epidemiological Trends of Racial Differences in Early- and Late-Onset Group B Streptococcus Disease in Tennessee.” *Clinical Infectious Diseases*, October. <https://doi.org/10.1093/cid/cia1511>.
 37. Kim, Ahra, Sarah L. Rader, Thomas V. Fernandez, Simon N. **Vandekar**, and Alan S. Lewis. 2020. “Leveraging Aggression Risk Gene Expression in the Developing and Adult Human Brain to Guide Future Precision Interventions.” *Molecular Psychiatry*, October, 1–3. <https://doi.org/10.1038/s41380-020-00903-3>.
 38. Avery, Suzanne N., Kristan Armstrong, Maureen McHugo, Simon **Vandekar**, Jennifer Urbano Blackford, Neil D. Woodward, and Stephan Heckers. 2021. “Relational Memory in the Early Stage of Psychosis: A 2-Year Follow-up Study.” *Schizophrenia Bulletin* 47 (1): 75–86.
 39. Connell, Jennifer, Ahra Kim, Nathan E. Brummel, Mayur B. Patel, Simon N. **Vandekar**, Pratik Pandharipande, Robert S. Dittus, Stephan Heckers, E. Wes Ely, and Jo Ellen Wilson. 2021. “Advanced Age Is Associated With Catatonia in Critical Illness: Results From the Delirium and Catatonia Prospective Cohort Investigation.” *Frontiers in Psychiatry* 12: 1999. <https://doi.org/10.3389/fpsy.2021.673166>.
 40. McHugo, Maureen, Suzanne Avery, Kristan Armstrong, Baxter P. Rogers, Simon N. **Vandekar**, Neil D. Woodward, Jennifer Urbano Blackford, and Stephan Heckers. 2021. “Anterior Hippocampal Dysfunction in Early Psychosis: A 2-Year Follow-up Study.” *Psychological Medicine*, 1–10.
 41. McKernan, Lindsey C., Leslie J. Crofford, Ahra Kim, Simon N. **Vandekar**, William S. Reynolds, Kathryn A. Hansen, Daniel J. Clauw, and David A. Williams. 2021. “Electronic Delivery of Pain Education for Chronic Overlapping Pain Conditions: A Prospective Cohort Study.” *Pain Medicine* 22 (10): 2252–62. <https://doi.org/10.1093/pm/pnab018>.
 42. Weinstein, Sarah M., Simon N. **Vandekar**, Azeez Adebimpe, Tinashe M. Tapera, Timothy Robert-Fitzgerald, Ruben C. Gur, Raquel E. Gur, et al. 2021. “A Simple Permutation-Based Test of Intermodal Correspondence.” *Human Brain Mapping* 42 (16): 5175–87. <https://doi.org/10.1002/hbm.25577>.
 43. Muscatello, Rachael A., Simon N. **Vandekar**, and Blythe A. Corbett. 2021. “Evidence for Decreased Parasympathetic Response to a Novel Peer Interaction in Older Children with Autism Spectrum Disorder: A Case-Control Study.” *Journal of Neurodevelopmental Disorders* 13 (1): 6. <https://doi.org/10.1186/s11689-020-09354-x>.
 44. Roeske, Maxwell J., Maureen McHugo, Simon **Vandekar**, Jennifer Urbano Blackford, Neil D. Woodward, and Stephan Heckers. 2021. “Incomplete Hippocampal Inversion in Schizophrenia: Prevalence, Severity, and Impact on

- Hippocampal Structure.” *Molecular Psychiatry*, January, 1–10.
<https://doi.org/10.1038/s41380-020-01010-z>.
45. **Vandekar**, Simon N., and Jeremy Stephens. 2021. “Improving the Replicability of Neuroimaging Findings by Thresholding Effect Sizes Instead of P-Values.” *Human Brain Mapping* 42 (8): 2393–98. <https://doi.org/10.1002/hbm.25374>.
 46. Muscatello, Rachael A., Ahra Kim, Simon **Vandekar**, and Blythe A. Corbett. 2021. “Diagnostic and Physical Effects in Parasympathetic Response to Social Evaluation in Youth With and Without Autism Spectrum Disorder.” *Journal of Autism and Developmental Disorders* 52 (8): 3427–42.
<https://doi.org/10.1007/s10803-021-05224-0>.
 47. Avery, Suzanne N., Anna S. Huang, Julia M. Sheffield, Baxter P. Rogers, Simon **Vandekar**, Alan Anticevic, and Neil D. Woodward. 2021. “Development of Thalamocortical Structural Connectivity in Typically Developing and Psychosis Spectrum Youth.” *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, October. <https://doi.org/10.1016/j.bpsc.2021.09.009>.
 48. Hamdan, Lubna, Varvara Probst, Zaid Haddadin, Herdi Rahman, Andrew J. Spieker, Simon **Vandekar**, Laura S. Stewart, et al. 2021. “Influenza Clinical Testing and Oseltamivir Treatment in Hospitalized Children with Acute Respiratory Illness, 2015–2016.” *Influenza and Other Respiratory Viruses*, October. <https://doi.org/10.1111/irv.12927>.
 49. Huang, Anna S., Baxter P. Rogers, Julia M. Sheffield, Simon **Vandekar**, Alan Anticevic, and Neil D. Woodward. 2021. “Characterizing Effects of Age, Sex and Psychosis Symptoms on Thalamocortical Functional Connectivity in Youth.” *NeuroImage* 243 (November): 118562.
<https://doi.org/10.1016/j.neuroimage.2021.118562>.
 50. Chen, Bob, Cherie’ R. Scurrah, Eliot T. McKinley, Alan J. Simmons, Marisol A. Ramirez-Solano, Xiangzhu Zhu, Nicholas O. Markham, et al. 2021. “Differential Pre-Malignant Programs and Microenvironment Chart Distinct Paths to Malignancy in Human Colorectal Polyps.” *Cell* 184 (26): 6262–6280.e26.
<https://doi.org/10.1016/j.cell.2021.11.031>.
 51. Baller, Erica B., Alessandra M. Valcarcel, Azeez Adebimpe, Aaron Alexander-Bloch, Zaixu Cui, Ruben C. Gur, Raquel E. Gur, Bart L. Larsen, Kristin A. Linn, and Carly M. O’Donnell. 2022. “Developmental Coupling of Cerebral Blood Flow and fMRI Fluctuations in Youth.” *Cell Reports* 38 (13): 110576.
 52. Corbett, Blythe A., Rachael A. Muscatello, Ahra Kim, Simon **Vandekar**, Sara Duffus, Sloane Sparks, and Yajas Tanguturi. 2022. “Examination of Pubertal Timing and Tempo in Females and Males with Autism Spectrum Disorder Compared to Typically Developing Youth.” *Autism Research* 15 (10): 1894–1908. <https://doi.org/10.1002/aur.2786>.
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Other publications:

1. **Vandekar**, Simon, 2019. Two Methodologies in “Amygdala Activation and Connectivity to Emotional Processing Distinguishes Asymptomatic Patients With Bipolar Disorders and Unipolar Depression” That Can Produce False-Positive Results and Some Statistical Recommendations. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*.
<https://doi.org/10.1016/j.bpsc.2018.12.007>
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3. Wrobel, Julia, Coleman Harris, and Simon **Vandekar**. 2023. “Statistical Analysis of Multiplex Immunofluorescence and Immunohistochemistry Imaging Data.” In *Statistical Genomics*, edited by Brooke Fridley and Xuefeng Wang, 141–68. Methods in Molecular Biology. New York, NY: Springer US.
https://doi.org/10.1007/978-1-0716-2986-4_8.

Invited Presentations:

1. “Multivariate Nonlinear Age Trajectories in Cortical Thickness,” 2013. Johns Hopkins University. Baltimore, MD.
2. “Analysis of Spatially Correlated Developmental Neuroimaging Data,” 2015. North Carolina State University. Raleigh, NC.
3. “More Powerful Permutation Tests Using Voxel-wise Transformations,” 2017. Statistical Methods in Imaging. Pittsburgh, PA. – Student paper award
4. “Powerful, Fast, and Robust Family-Wise Error Control for Neuroimaging,” 2017. JSM. Baltimore, MD.
5. “Association Tests for Neuroimaging Studies of Development and Disease,” 2018. Johns Hopkins University. Baltimore, MD.
6. “Association Tests for Neuroimaging Studies of Development and Disease,” 2018. Columbia University. New York, NY.
7. “Robust Spatial Extent Inference with a Semiparametric Bootstrap Joint Testing Procedure,” 2019. ENAR Conference. Philadelphia, PA.
8. “Robust Spatial Extent Inference with a Semiparametric Bootstrap Joint Testing Procedure,” 2019. Statistical Methods in Imaging. Irvine, CA.
9. “On Testing the Spatial Correspondence Between Maps of Human Brain Structure and Function,” 2019. International Chinese Statistical Association. Raleigh, NC.
10. “Robust Spatial Extent Inference in R using the pbj Package,” 2019. Department of Statistics, Indiana University, Bloomington. Bloomington, IN.
11. “Robust Spatial Extent Inference in R using the pbj Package,” 2019. Department of Biostatistics, Boston University. Boston, MA.

12. "Robust Spatial Extent Inference in R using the pbj Package," 2019. Department of Biomedical Informatics, University of Arkansas for Medical Sciences. Little Rock, AR.
13. "Robust Spatial Extent Inference in R using the pbj Package," 2020. Department of Epidemiology and Biostatistics, University of San Francisco. San Francisco, CA.
14. "Thresholding Statistical Maps by Effect Sizes for Spatial Extent Inference of Neuroimages," 2020. CM Statistics Conference. London, UK.
15. "Robust spatial extent inference in R using the pbj package," 2021. Department of Biostatistics, University of California, San Francisco. Virtual.
16. "A general measure of effect size for neuroimaging analysis," 2021. Penn Statistical Imaging Endeavor, University of Pennsylvania. Virtual.
17. "A general measure of effect size for neuroimaging analysis," 2021. Department of Biostatistics, Emory University. Virtual.
18. "A robust measure of effect size for neuroimage analysis," 2021. CM Statistics Conference. London, UK.
19. "Cell Normalization and Phenotyping for Multiplexed Single-Cell Imaging," 2022. ENAR Conference. Houston, TX.
20. "Cell Normalization and Phenotyping for Multiplexed Single-Cell Imaging," 2022. JSM. Washington, DC.
21. "Cell Normalization and Phenotyping for Multiplexed Single-Cell Imaging," 2022. Jefferson University, Philadelphia, PA.
22. "Robust and reproducible group-level neuroimage analysis in R with the pbj package," 2022. CM Statistics Conference. London, UK.
23. "Effect Sizes and Replicability in Brain-wide Association Studies (BWAS)," 2023. Computational and Methodological (CM) Statistics Conference. Berlin, Germany.
24. "Normalization and Cell Phenotyping for mIF Data," 2024. Pacific Symposium on Biocomputing. Kona, HI.
25. "Effect sizes and replicability in longitudinal studies of brain-phenotype associations," 2024. Joint Statistical Meetings (JSM). Portland, OR.